

Under-Graduate Project

1 From Lagrangian to Hamiltonian Mechanics

The position of a mechanical system with d degrees of freedom is typically described by a vector of generalized coordinates $q = (q_1, \dots, q_d)^T$. The dynamics of such a system are built upon two fundamental quantities: the kinetic energy $T = T(q, \dot{q})$ and the potential energy $U = U(q)$. By taking the difference of these energies, we construct the Lagrangian of the system:

$$L(q, \dot{q}) = T(q, \dot{q}) - U(q) \quad (1)$$

The motion of the system is then determined by the Euler-Lagrange equations:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = \frac{\partial L}{\partial q} \quad (2)$$

While these equations successfully model a vast array of physical systems, they inherently constitute a system of second-order differential equations.

To simplify the structure and transform this framework into a highly symmetric system of first-order differential equations, we introduce a new set of independent variables called Poisson's variables, or the conjugate momenta, defined as:

$$p_k = \frac{\partial L}{\partial \dot{q}_k}(q, \dot{q}) \quad \text{for } k = 1, \dots, d \quad (3)$$

This transition mathematically relies on the Legendre transform. Specifically, it is required that the definition of the momenta establishes a continuously differentiable bijection $\dot{q} \leftrightarrow p$ between the velocities $\dot{q}$ and the momenta p for every configuration q . This critical requirement ensures that the relationship is invertible, allowing us to explicitly express the velocities as a function of the coordinates and momenta, $\dot{q} = \dot{q}(p, q)$.

With this inversion established, we can rigorously define the Hamiltonian function H as a function of p and q by substituting $\dot{q}(p, q)$ into the transformed energy equation:

$$H(p, q) = p^T \dot{q}(p, q) - L(q, \dot{q}(p, q)) \quad (4)$$

Theorem 1.3. *Lagrange's equations are equivalent to Hamilton's canonical equations:*

$$\dot{p}_k = -\frac{\partial H}{\partial q_k}, \quad \dot{q}_k = \frac{\partial H}{\partial p_k} \quad (5)$$

Proof. By definition, the Hamiltonian is a function of p and q , meaning we must treat $\dot{q}$ as a function of p and q . When we take the partial derivatives of H with respect to p and q using the product rule and chain rule, we obtain:

$$\frac{\partial H}{\partial p} = \dot{q}^T + p^T \frac{\partial \dot{q}}{\partial p} - \frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial p} = \dot{q}^T \quad (6)$$

$$\frac{\partial H}{\partial q} = p^T \frac{\partial \dot{q}}{\partial q} - \frac{\partial L}{\partial q} - \frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial q} = -\frac{\partial L}{\partial q} \quad (7)$$

Notice how the terms $p^T \frac{\partial \dot{q}}{\partial p}$ and $\frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial p}$ perfectly cancel each other out precisely because we defined our momenta as $p = \frac{\partial L}{\partial \dot{q}}$. The exact same cancellation occurs in the second equation.

To fully understand why this establishes equivalence, we must look at the theoretical implications of these two resulting derivatives.

The first result, $\frac{\partial H}{\partial p} = \dot{q}^T$ (or simply $\dot{q} = \frac{\partial H}{\partial p}$), is a kinematic relation. It inherently satisfies the definition of our conjugate momenta and ensures that our variable substitution via the Legendre transform remains consistent.

The second result yields $\frac{\partial H}{\partial q} = -\frac{\partial L}{\partial q}$. If we substitute the original second-order Euler-Lagrange equation, $\frac{\partial L}{\partial q} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right)$, and recall our momenta definition $p = \frac{\partial L}{\partial \dot{q}}$, the right side becomes precisely $-\dot{p}$. Therefore, we arrive at $\dot{p} = -\frac{\partial H}{\partial q}$.

Together, these steps demonstrate that the original, singular second-order Lagrangian system is perfectly decoupled into the symmetric pair of first-order Hamiltonian equations.

Case of Quadratic Kinetic Energy (T)

In many classical mechanical systems, the kinetic energy is a quadratic function of the velocities. Specifically, suppose that $T = \frac{1}{2} \dot{q}^T M(q) \dot{q}$, where the mass matrix $M(q)$ is a symmetric and positive definite matrix.

Recalling that the Lagrangian is $L(q, \dot{q}) = T(q, \dot{q}) - U(q)$, and noting that the potential energy $U(q)$ is independent of velocity, the conjugate momentum is computed as the gradient of the kinetic energy with respect to $\dot{q}$. Because $M(q)$ is symmetric, this yields:

$$p = \nabla_{\dot{q}} L = M(q) \dot{q} \quad (8)$$

For a fixed configuration q , the positive definiteness of $M(q)$ ensures that this matrix is non-singular and invertible. Consequently, this relationship strictly satisfies the conditions for the existence of the Legendre transform. We can explicitly invert this relation to solve for the velocities in terms of the momenta:

$$\dot{q} = M(q)^{-1} p \quad (9)$$

To construct the Hamiltonian $H(p, q)$, we substitute this inverted velocity relation into the fundamental definition $H(p, q) = p^T \dot{q} - L(q, \dot{q})$:

$$H(p, q) = p^T M(q)^{-1} p - L(q, M(q)^{-1} p) \quad (10)$$

Expanding the Lagrangian term $L = T - U$ using our momentum substitution:

$$H(p, q) = p^T M(q)^{-1} p - \left(\frac{1}{2} (M(q)^{-1} p)^T M(q) (M(q)^{-1} p) - U(q) \right) \quad (11)$$

$$= p^T M(q)^{-1} p - \frac{1}{2} p^T (M(q)^{-1})^T M(q) M(q)^{-1} p + U(q) \quad (12)$$

Because $M(q)$ is symmetric, its inverse $M(q)^{-1}$ is also symmetric, meaning $(M(q)^{-1})^T = M(q)^{-1}$. Since $M(q)M(q)^{-1} = I$, the kinetic energy term simplifies cleanly:

$$H(p, q) = p^T M(q)^{-1} p - \frac{1}{2} p^T M(q)^{-1} p + U(q) \quad (13)$$

$$= \frac{1}{2} p^T M(q)^{-1} p + U(q) \quad (14)$$

Recognizing that the first term is exactly the kinetic energy T expressed in terms of momentum, we arrive at the fundamental physical interpretation:

$$H = T + U \quad (15)$$

Therefore, for systems with quadratic kinetic energy, the Hamiltonian is identically the total energy of the mechanical system.

Example: The Mathematical Pendulum

To illustrate this transition physically and mathematically, consider the simple mathematical pendulum. We define its position using a single generalized coordinate, the angle $q = \alpha$.

The kinetic energy T and potential energy U of the pendulum are given by:

$$T = \frac{1}{2} m l^2 \dot{q}^2, \quad U = -mgl \cos q \quad (16)$$

where m is the mass, l is the length of the pendulum, and g is the acceleration due to gravity. The Lagrangian $L = T - U$ is therefore:

$$L(q, \dot{q}) = \frac{1}{2} m l^2 \dot{q}^2 + mgl \cos q \quad (17)$$

Applying the Euler-Lagrange equation, $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = \frac{\partial L}{\partial q}$, we obtain the second-order differential equation of motion:

$$m l^2 \ddot{q} = -mgl \sin q \implies \ddot{q} + \frac{g}{l} \sin q = 0 \quad (18)$$

This is the classic formulation of the pendulum's dynamics. To transform this into a Hamiltonian system, we first simplify the expressions by applying the normalization $m = l = g = 1$. Our normalized Lagrangian is $L = \frac{1}{2} \dot{q}^2 + \cos q$.

We introduce the conjugate momentum p , which represents the angular momentum:

$$p = \frac{\partial L}{\partial \dot{q}} = \dot{q} \quad (19)$$

Using the Legendre transform to eliminate $\dot{q}$, we construct the Hamiltonian function $H(p, q)$:

$$H(p, q) = p\dot{q} - L = p(p) - \left(\frac{1}{2}p^2 + \cos q \right) = \frac{1}{2}p^2 - \cos q \quad (20)$$

Applying Hamilton's canonical equations, we recover the decoupled first-order system:

$$\dot{q} = \frac{\partial H}{\partial p} = p, \quad \dot{p} = -\frac{\partial H}{\partial q} = -\sin q \quad (21)$$

Physically, because the original kinetic energy T was quadratic with respect to the velocities, the resulting Hamiltonian $H = T + U$ represents the total mechanical energy of the system. The Hamiltonian formulation thus shifts our perspective: rather than viewing the pendulum's motion merely as a second-order response to restoring forces (the Lagrangian view), we now understand its evolution as a geometric flow through phase space (p, q) that strictly conserves the total energy, $H(p, q)$.

2 Symplectic Transformations

2.1 Oriented Area

To understand the geometric properties of Hamiltonian systems, we first consider a two-dimensional parallelogram P in phase space spanned by two vectors $\xi = (\xi^p, \xi^q)^T$ and $\eta = (\eta^p, \eta^q)^T$.

In one dimension ($d = 1$), the oriented area of this parallelogram is simply the determinant of the coordinates:

$$\text{or.area}(P) = \det \begin{pmatrix} \xi^p & \eta^p \\ \xi^q & \eta^q \end{pmatrix} = \xi^p \eta^q - \xi^q \eta^p \quad (22)$$

For higher dimensions ($d > 1$), we generalize this by taking the sum of the oriented areas of the projections of the parallelogram onto the coordinate planes (p_i, q_i) . This is expressed using the bilinear mapping $\omega(\xi, \eta)$:

$$\omega(\xi, \eta) := \sum_{i=1}^d \det \begin{pmatrix} \xi_i^p & \eta_i^p \\ \xi_i^q & \eta_i^q \end{pmatrix} = \sum_{i=1}^d (\xi_i^p \eta_i^q - \xi_i^q \eta_i^p) \quad (23)$$

In matrix notation, this takes the elegant form $\omega(\xi, \eta) = \xi^T J \eta$, where $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$ and I is the $d \times d$ identity matrix.

2.2 Defining Symplectic Maps

Definition 2.1. A linear mapping $A : \mathbb{R}^{2d} \rightarrow \mathbb{R}^{2d}$ is called symplectic if

$$A^T J A = J \quad (24)$$

or, equivalently, if $\omega(A\xi, A\eta) = \omega(\xi, \eta)$ for all $\xi, \eta \in \mathbb{R}^{2d}$.

Example: 2D Phase Space and Area Preservation

To illustrate this definition in the simplest case, consider a 2D phase space ($d = 1$) with coordinates (p, q) . A linear transformation A on this space is symplectic if and only if $\det(A) = 1$.

Proof. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a general 2×2 transformation matrix, and

recall the standard symplectic structure matrix for $d = 1$ is $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. We

evaluate the left-hand side of the symplecticity condition, $A^T J A$, step-by-step. First, computing the product JA :

$$JA = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} c & d \\ -a & -b \end{pmatrix} \quad (25)$$

Next, multiplying this result by A^T from the left yields:

$$A^T JA = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} c & d \\ -a & -b \end{pmatrix} = \begin{pmatrix} ac - ac & ad - bc \\ bc - ad & bd - bd \end{pmatrix} \quad (26)$$

Simplifying the diagonal terms, we obtain:

$$A^T JA = \begin{pmatrix} 0 & ad - bc \\ -(ad - bc) & 0 \end{pmatrix} \quad (27)$$

Recognizing that the determinant of A is precisely $\det(A) = ad - bc$, we can factor this scalar value out of the matrix:

$$A^T JA = \det(A) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \det(A) J \quad (28)$$

We now substitute this universal 2×2 identity directly into our defining symplecticity condition:

$$A^T JA = J \iff \det(A) J = J \quad (29)$$

Since the structure matrix J is non-zero, we can equate the scalar coefficients. This equation holds true if and only if $\det(A) = 1$.

This concept extends to non-linear maps by examining their local linear approximations (their Jacobians).

Definition 2.2. A differentiable map $g : U \rightarrow \mathbb{R}^{2d}$ (where $U \subset \mathbb{R}^{2d}$ is an open set) is called symplectic if the Jacobian matrix $g'(p, q)$ is everywhere symplectic, i.e., if

$$g'(p, q)^T J g'(p, q) = J \quad \text{or} \quad \omega(g'(p, q)\xi, g'(p, q)\eta) = \omega(\xi, \eta) \quad (30)$$

2.3 Geometric Interpretation on Manifolds

We can geometrically interpret this by considering a 2-dimensional sub-manifold M defined as the image $M = \psi(K)$ of a compact set $K \subset \mathbb{R}^2$, where $\psi(s, t)$ is a continuously differentiable parameterization.

If we divide the compact set K into infinitesimal compartments $ds dt$, the manifold M can be approximated as the union of small parallelograms spanned by the tangent vectors $\frac{\partial \psi}{\partial s}(s, t) ds$ and $\frac{\partial \psi}{\partial t}(s, t) dt$. Summing the oriented areas of the projections of all these infinitesimal parallelograms yields the total oriented area over the manifold:

$$\Omega(M) = \iint_K \omega \left(\frac{\partial \psi}{\partial s}(s, t), \frac{\partial \psi}{\partial t}(s, t) \right) ds dt \quad (31)$$

Lemma 2.3. *If the mapping $g : U \rightarrow \mathbb{R}^{2d}$ is symplectic on U , then it preserves the expression $\Omega(M)$, i.e.,*

$$\Omega(g(M)) = \Omega(M) \quad (32)$$

holds for all 2-dimensional manifolds M that can be represented as the image of a continuously differentiable function ψ .

Proof. The transformed manifold $g(M)$ can be parameterized by the composition $g \circ \psi$. Setting up the integral for the area of this new manifold, we have:

$$\Omega(g(M)) = \iint_K \omega \left(\frac{\partial(g \circ \psi)}{\partial s}(s, t), \frac{\partial(g \circ \psi)}{\partial t}(s, t) \right) ds dt \quad (33)$$

By the chain rule, the partial derivatives are $\frac{\partial}{\partial s}(g \circ \psi) = g'(\psi(s, t)) \frac{\partial \psi}{\partial s}$ (and identically for t). Because g is a symplectic transformation, its Jacobian g' preserves the bilinear map ω , meaning $\omega(g'\xi, g'\eta) = \omega(\xi, \eta)$. Therefore, the integrand simplifies entirely, yielding:

$$\Omega(g(M)) = \iint_K \omega \left(\frac{\partial \psi}{\partial s}(s, t), \frac{\partial \psi}{\partial t}(s, t) \right) ds dt = \Omega(M) \quad (34)$$

This confirms that any symplectic mapping (including non-linear ones) is fundamentally area-preserving.

3 Theorems on Symplectic Flow

Theorem 2.4 (Poincaré 1899). *Let $H(p, q)$ be a twice continuously differentiable function on $U \subset \mathbb{R}^{2d}$. Then, for each fixed t , the flow φ_t is a symplectic transformation wherever it is defined.*

Proof. Let $y = (p, q)$. We write the Hamiltonian system in the form $\dot{y} = J^{-1} \nabla H(y)$. The flow is given by $\varphi_t(y_0) = y(t)$.

We first derive the variational equation by differentiating the Jacobian of the flow, $\Psi = \frac{\partial \varphi_t}{\partial y_0}$, with respect to t :

$$\frac{d}{dt} \left(\frac{\partial \varphi_t}{\partial y_0} \right) = \frac{\partial}{\partial y_0} \left(\frac{d\varphi_t}{dt} \right) = \frac{\partial}{\partial y_0} (J^{-1} \nabla H(\varphi_t(y_0))) \quad (35)$$

Applying the chain rule yields:

$$\frac{d}{dt} \left(\frac{\partial \varphi_t}{\partial y_0} \right) = J^{-1} \nabla^2 H(\varphi_t(y_0)) \frac{\partial \varphi_t}{\partial y_0} \quad (36)$$

where $\nabla^2 H$ is the symmetric Hessian matrix of H . Thus, Ψ is a solution to the variational equation $\dot{\Psi} = J^{-1} \nabla^2 H(\varphi_t(y_0)) \Psi$.

To show the flow is symplectic, we differentiate the condition $\Psi^T J \Psi$ with respect to time:

$$\frac{d}{dt} (\Psi^T J \Psi) = \dot{\Psi}^T J \Psi + \Psi^T J \dot{\Psi} \quad (37)$$

Substituting our derived variational equation gives:

$$= (J^{-1} \nabla^2 H(\varphi_t(y_0)) \Psi)^T J \Psi + \Psi^T J (J^{-1} \nabla^2 H(\varphi_t(y_0)) \Psi) \quad (38)$$

Because $J^T = -J$ and $J^{-T} J = -I$, this simplifies to:

$$= \Psi^T \nabla^2 H(\varphi_t(y_0)) J^{-T} J \Psi + \Psi^T \nabla^2 H(\varphi_t(y_0)) \Psi = 0 \quad (39)$$

Since $\frac{d}{dt} (\Psi^T J \Psi) = 0$, the expression $\Psi^T J \Psi$ is constant for all t . At $t = 0$, φ_0 is the identity map, meaning $\Psi(0) = I$, which gives $I^T J I = J$. Therefore, $\left(\frac{\partial \varphi_t}{\partial y_0} \right)^T J \left(\frac{\partial \varphi_t}{\partial y_0} \right) = J$ for all t , proving the flow is symplectic.

Definition. A differential equation $\dot{y} = f(y)$ is locally Hamiltonian if, for every $y_0 \in U$, there exists a neighbourhood where $f(y) = J^{-1} \nabla H(y)$ for some function H .

Theorem 2.6. Let $f : U \rightarrow \mathbb{R}^{2d}$ be continuously differentiable. Then, $\dot{y} = f(y)$ is locally Hamiltonian if and only if its flow $\varphi_t(y)$ is symplectic for all $y \in U$ and for all sufficiently small t .

Proof. The necessity follows directly from Theorem 2.4. To prove sufficiency, we assume the flow φ_t is symplectic and must prove the local existence of a function $H(y)$ such that $f(y) = J^{-1} \nabla H(y)$.

We start by differentiating the symplecticity condition, $\left(\frac{\partial \varphi_t}{\partial y_0} \right)^T J \left(\frac{\partial \varphi_t}{\partial y_0} \right) = J$, with respect to t , knowing that $\frac{\partial \varphi_t}{\partial y_0}$ solves the variational equation $\dot{\Psi} = f'(\varphi_t(y_0)) \Psi$:

$$\frac{d}{dt} \left(\left(\frac{\partial \varphi_t}{\partial y_0} \right)^T J \left(\frac{\partial \varphi_t}{\partial y_0} \right) \right) = \left(\frac{\partial \varphi_t}{\partial y_0} \right)^T (f'(\varphi_t(y_0))^T J + J f'(\varphi_t(y_0))) \left(\frac{\partial \varphi_t}{\partial y_0} \right) = 0 \quad (40)$$

Putting $t = 0$, where $\varphi_0(y_0) = y_0$ and $\frac{\partial \varphi_0}{\partial y_0} = I$, it follows from $J = -J^T$ that $Jf'(y_0)$ is a symmetric matrix for all y_0 .

To find H , let us assume $y_0 = 0$ and consider a ball around 0 contained in U . On this ball, we construct the function $H(y)$ as:

$$H(y) = \int_0^1 y^T Jf(ty) dt + \text{Const} \quad (41)$$

Let $g(y) = Jf(y)$; since its Jacobian $g'(y) = Jf'(y)$ is symmetric, we have $\frac{\partial g_i}{\partial y_k} = \frac{\partial g_k}{\partial y_i}$. Differentiating $H(y)$ with respect to y_k yields:

$$\frac{\partial H}{\partial y_k}(y) = \int_0^1 \left(g_k(ty) + y^T \frac{\partial g}{\partial y_k}(ty)t \right) dt \quad (42)$$

Using the symmetry property, the integrand simplifies as an exact derivative:

$$\frac{\partial H}{\partial y_k}(y) = \int_0^1 \frac{d}{dt} (tg_k(ty)) dt = g_k(y) \quad (43)$$

This proves that $\nabla H(y) = g(y) = Jf(y)$. Rearranging gives $f(y) = J^{-1}\nabla H(y)$, concluding the proof that the system is locally Hamiltonian.

4 Symplecticity of the Symplectic Euler Method

Definition 3.1. *A numerical one-step method is called symplectic if the one-step map $y_1 = \Phi_h(y_0)$ is symplectic whenever the method is applied to a smooth Hamiltonian system*

To prove that the Symplectic Euler method is indeed symplectic, we examine the Jacobian of the transformation. The method is defined by the following step equations:

$$p_{n+1} = p_n - hH_q(p_{n+1}, q_n)$$

$$q_{n+1} = q_n + hH_p(p_{n+1}, q_n)$$

4.1 Step 1: Implicit Differentiation

We first differentiate both equations with respect to the initial states p_n and q_n .

Block A: Differentiating $p_{n+1} = p_n - hH_q(p_{n+1}, q_n)$

1. With respect to p_n :

$$\frac{\partial p_{n+1}}{\partial p_n} = I - hH_{qp}^T \frac{\partial p_{n+1}}{\partial p_n} - hH_{qq} \frac{\partial q_n}{\partial p_n}$$

Since $\frac{\partial q_n}{\partial p_n} = 0$, we get:

$$(I + hH_{qp}^T) \frac{\partial p_{n+1}}{\partial p_n} = I \quad (3)$$

2. With respect to q_n :

$$\frac{\partial p_{n+1}}{\partial q_n} = 0 - hH_{qp}^T \frac{\partial p_{n+1}}{\partial q_n} - hH_{qq} \frac{\partial q_n}{\partial q_n}$$

Since $\frac{\partial q_n}{\partial q_n} = I$, we get:

$$(I + hH_{qp}^T) \frac{\partial p_{n+1}}{\partial q_n} = -hH_{qq} \quad (4)$$

Block B: Differentiating $q_{n+1} = q_n + hH_p(p_{n+1}, q_n)$

1. With respect to p_n :

$$\frac{\partial q_{n+1}}{\partial p_n} = \frac{\partial q_n}{\partial p_n} + hH_{pp} \frac{\partial p_{n+1}}{\partial p_n} + hH_{qp} \frac{\partial q_n}{\partial p_n}$$

Since $\frac{\partial q_n}{\partial p_n} = 0$, we get:

$$-hH_{pp} \frac{\partial p_{n+1}}{\partial p_n} + \frac{\partial q_{n+1}}{\partial p_n} = 0 \quad (5)$$

2. With respect to q_n :

$$\frac{\partial q_{n+1}}{\partial q_n} = I + hH_{pp} \frac{\partial p_{n+1}}{\partial q_n} + hH_{qp} \frac{\partial q_n}{\partial q_n}$$

Since $\frac{\partial q_n}{\partial q_n} = I$, we get:

$$-hH_{pp} \frac{\partial p_{n+1}}{\partial q_n} + \frac{\partial q_{n+1}}{\partial q_n} = I + hH_{qp} \quad (6)$$

4.2 Step 2: The Compact Block Matrix Formulation

Combining equations (3), (4), (5), and (6) into a single block matrix system yields:

$$\begin{pmatrix} I + hH_{qp}^T & 0 \\ -hH_{pp} & I \end{pmatrix} \begin{pmatrix} \frac{\partial p_{n+1}}{\partial p_n} & \frac{\partial p_{n+1}}{\partial q_n} \\ \frac{\partial q_{n+1}}{\partial p_n} & \frac{\partial q_{n+1}}{\partial q_n} \end{pmatrix} = \begin{pmatrix} I & -hH_{qq} \\ 0 & I + hH_{qp} \end{pmatrix}$$

Let us define the helper matrices A and B :

$$A = \begin{pmatrix} I + hH_{qp}^T & 0 \\ -hH_{pp} & I \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} I & -hH_{qq} \\ 0 & I + hH_{qp} \end{pmatrix}$$

Thus, we can write the relation compactly as:

$$AG = B \implies G = A^{-1}B \quad (7)$$

4.3 Step 3: Proving Symplecticity using Transposes

In symplectic linear algebra, a matrix G is symplectic ($G^T J G = J$) if and only if its transpose is also symplectic ($G J G^T = J$). By using this equivalent definition, the proof collapses into three lines:

1. Substitute $G = A^{-1}B$ and $G^T = B^T A^{-T}$:

$$A^{-1}B J B^T A^{-T} = J$$

2. Multiply on the left by A :

$$A(A^{-1}B J B^T A^{-T}) = A J \implies B J B^T A^{-T} = A J$$

3. Multiply on the right by A^T :

$$(B J B^T A^{-T}) A^T = A J A^T \implies B J B^T = A J A^T$$

This directly proves that showing $A J A^T = B J B^T$ is equivalent to showing the method is symplectic.

Part A: Calculating $A J A^T$ Let $X = I + h H_{qp}^T$ and $Y = -h H_{pp}$ (where $Y^T = Y$ due to symmetry).

$$A = \begin{pmatrix} X & 0 \\ Y & I \end{pmatrix}$$

$$A J = \begin{pmatrix} X & 0 \\ Y & I \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} = \begin{pmatrix} 0 & X \\ -I & Y \end{pmatrix}$$

$$A J A^T = \begin{pmatrix} 0 & X \\ -I & Y \end{pmatrix} \begin{pmatrix} X^T & Y^T \\ 0 & I \end{pmatrix} = \begin{pmatrix} 0 & X \\ -X^T & -Y^T + Y \end{pmatrix}$$

Since Y is symmetric, the bottom-right term $-Y^T + Y = 0$. Substituting X back in yields:

$$A J A^T = \begin{pmatrix} 0 & I + h H_{qp}^T \\ -(I + h H_{qp}) & 0 \end{pmatrix} \quad (9)$$

Part B: Calculating $B J B^T$ Let $Z = -h H_{qq}$ (where $Z^T = Z$ due to symmetry) and $W = I + h H_{qp}$.

$$B = \begin{pmatrix} I & Z \\ 0 & W \end{pmatrix}$$

$$B J = \begin{pmatrix} I & Z \\ 0 & W \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} = \begin{pmatrix} -Z & I \\ -W & 0 \end{pmatrix}$$

$$B J B^T = \begin{pmatrix} -Z & I \\ -W & 0 \end{pmatrix} \begin{pmatrix} I & 0 \\ Z^T & W^T \end{pmatrix} = \begin{pmatrix} -Z + Z^T & W^T \\ -W & 0 \end{pmatrix}$$

Since Z is symmetric, the top-left term $-Z + Z^T = 0$. Substituting W back in yields:

$$BJB^T = \begin{pmatrix} 0 & I + hH_{qp}^T \\ -(I + hH_{qp}) & 0 \end{pmatrix} \quad (10)$$

Comparing Equations (9) and (10), $AJA^T = BJB^T$. Therefore, the Symplectic Euler method is exactly symplectic.

5 Theorem 3.4 (The Implicit Midpoint Rule)

The implicit midpoint rule:

$$y_{n+1} = y_n + hJ^{-1}\nabla H\left(\frac{y_{n+1} + y_n}{2}\right) \quad (1)$$

is a symplectic method of order 2.

5.1 Proof of Symplecticity

Let the state vector be $y = (p, q)^T \in \mathbb{R}^{2d}$, and let J be the standard skew-symmetric structure matrix:

$$J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$$

which satisfies the standard identities $J^T = -J$, $J^2 = -I$, and $J^{-1} = -J$.

We define the intermediate midpoint state as $y_{n+1/2} = \frac{y_{n+1} + y_n}{2}$ and the symmetric Hessian matrix of the Hamiltonian evaluated at this midpoint as:

$$S = \nabla^2 H(y_{n+1/2}) \quad \text{with} \quad S^T = S$$

Step 1: Implicit Differentiation To find the Jacobian matrix of the numerical step $G = \frac{\partial y_{n+1}}{\partial y_n}$, we differentiate both sides of Equation (1) with respect to the initial state y_n :

$$\frac{\partial y_{n+1}}{\partial y_n} = \frac{\partial y_n}{\partial y_n} + hJ^{-1} \frac{\partial}{\partial y_n} [\nabla H(y_{n+1/2})]$$

Applying the multivariable chain rule to the gradient term yields:

$$\frac{\partial y_{n+1}}{\partial y_n} = I + hJ^{-1} \left(\nabla^2 H(y_{n+1/2}) \cdot \frac{\partial y_{n+1/2}}{\partial y_n} \right)$$

$$G = I + hJ^{-1}S \left[\frac{1}{2} \left(\frac{\partial y_{n+1}}{\partial y_n} + \frac{\partial y_n}{\partial y_n} \right) \right]$$

$$G = I + \frac{h}{2}J^{-1}S(G + I)$$

Step 2: Isolating the Jacobian G Grouping the terms containing G on the left-hand side gives:

$$G - \frac{h}{2}J^{-1}SG = I + \frac{h}{2}J^{-1}S$$

$$\left(I - \frac{h}{2}J^{-1}S\right)G = I + \frac{h}{2}J^{-1}S \quad (2)$$

Let us define the helper matrices A and B as:

$$A = I - \frac{h}{2}J^{-1}S \quad \text{and} \quad B = I + \frac{h}{2}J^{-1}S$$

This simplifies our algebraic relation to $AG = B$, which means the Jacobian is given by $G = A^{-1}B$.

Step 3: Proving $G^T J G = J$ To show that the Jacobian G is symplectic, we must prove that $G^T J G = J$. Substituting $G = A^{-1}B$, we can rewrite this condition as:

$$(A^{-1}B)^T J (A^{-1}B) = J \implies B^T A^{-T} J A^{-1} B = J$$

Multiplying by A^T on the left and A on the right eliminates the inverse matrices without making any illegal commutation assumptions:

$$B^T J B = A^T J A \quad (3)$$

Let us expand both sides of Equation (3) to prove they are identical:

1. Expanding the Left-Hand Side ($A^T J A$): Using the property $J^{-1} = -J$, we write $A = I + \frac{h}{2}JS$. Its transpose is:

$$A^T = \left(I + \frac{h}{2}JS\right)^T = I + \frac{h}{2}S^T J^T = I - \frac{h}{2}SJ$$

Now, we compute the product $A^T J A$:

$$A^T J A = \left(I - \frac{h}{2}SJ\right) J \left(I + \frac{h}{2}JS\right)$$

$$A^T J A = \left(J - \frac{h}{2}SJ^2\right) \left(I + \frac{h}{2}JS\right)$$

Since $J^2 = -I$, this simplifies to:

$$A^T J A = \left(J + \frac{h}{2}S\right) \left(I + \frac{h}{2}JS\right)$$

$$A^T J A = J + \frac{h}{2}J^2 S + \frac{h}{2}S + \frac{h^2}{4}SJS$$

Using $J^2 = -I$ once more:

$$A^T J A = J - \frac{h}{2} S + \frac{h}{2} S + \frac{h^2}{4} S J S = J + \frac{h^2}{4} S J S \quad (4)$$

2. Expanding the Right-Hand Side ($B^T J B$): Similarly, we write $B = I - \frac{h}{2} J S$. Its transpose is:

$$B^T = \left(I - \frac{h}{2} J S \right)^T = I - \frac{h}{2} S^T J^T = I + \frac{h}{2} S J$$

Computing $B^T J B$:

$$\begin{aligned} B^T J B &= \left(I + \frac{h}{2} S J \right) J \left(I - \frac{h}{2} J S \right) \\ B^T J B &= \left(J + \frac{h}{2} S J^2 \right) \left(I - \frac{h}{2} J S \right) = \left(J - \frac{h}{2} S \right) \left(I - \frac{h}{2} J S \right) \\ B^T J B &= J - \frac{h}{2} J^2 S - \frac{h}{2} S + \frac{h^2}{4} S J S \end{aligned}$$

Using $J^2 = -I$:

$$B^T J B = J + \frac{h}{2} S - \frac{h}{2} S + \frac{h^2}{4} S J S = J + \frac{h^2}{4} S J S \quad (5)$$

Conclusion Comparing Equation (4) and Equation (5), we have proven rigorously that:

$$A^T J A = B^T J B = J + \frac{h^2}{4} S J S$$

Because this identity holds without requiring any commutation of S and J , the Jacobian $G = A^{-1} B$ is mathematically guaranteed to satisfy the symplecticity condition $G^T J G = J$ exactly. This completes the proof of Theorem 3.4.

6 The Störmer-Verlet Scheme

The Störmer-Verlet scheme is one of the most prominent numerical methods in physics, particularly designed for a specific class of second-order ordinary differential equations where the acceleration depends strictly on position:

$$\ddot{q} = f(q) \quad (44)$$

To analyze this geometrically, we transform the single second-order equation into a system of first-order equations. By defining an auxiliary velocity variable $v = \dot{q}$, we can substitute the original equation into its derivative to obtain the coupled system:

$$\dot{q} = v, \quad \dot{v} = f(q) \quad (45)$$

This decomposition directly mirrors the Hamiltonian formulation $\dot{q} = p, \dot{p} = f(q)$ for systems with normalized mass matrices.

6.1 The Classical 3-Term Recurrence and the Startup Problem

Historically, the Störmer-Verlet method was discretized by directly approximating the second derivative using a central difference formula. This yields the classical three-term recurrence relation:

$$q_{n+1} - 2q_n + q_{n-1} = h^2 f(q_n) \quad (1) \quad (46)$$

While computationally cheap because it does not require storing velocities, this recurrence suffers from a fundamental "startup problem". In physical mechanics, an initial value problem provides the initial position $q(0) = q_0$ and initial velocity $v(0) = v_0$. However, computing the very first step q_1 (at $n = 0$) requires the state q_{-1} :

$$q_1 - 2q_0 + q_{-1} = h^2 f(q_0) \quad (2) \quad (47)$$

The term q_{-1} represents a non-existent "ghost step" at time $t = -h$, meaning the classical method cannot begin on its own.

6.2 Deriving the Preliminary Starting Condition

To resolve the missing q_{-1} state, we utilize the given initial velocity v_0 . We approximate v_0 using a central difference over the interval $[-h, h]$:

$$v_0 = \frac{q_1 - q_{-1}}{2h} \quad (3) \quad (48)$$

We can solve Equation (3) for the ghost step:

$$q_{-1} = q_1 - 2hv_0 \quad (49)$$

Substituting this expression back into the initial recurrence (Equation 2) eliminates the unknown past state:

$$q_1 - 2q_0 + (q_1 - 2hv_0) = h^2 f(q_0) \quad (50)$$

Grouping the terms yields:

$$2q_1 - 2q_0 - 2hv_0 = h^2 f(q_0) \quad (51)$$

Dividing the entire equation by 2 and isolating q_1 provides the exact mathematical starting condition required to initiate the simulation:

$$q_1 = q_0 + hv_0 + \frac{h^2}{2} f(q_0) \quad (52)$$

6.3 The Explicit One-Step Formulation (Velocity Verlet)

The classical multi-step formulation poses practical issues beyond the startup calculation; primarily, it requires approximating velocities externally if they are needed to compute kinetic energy or check momentum conservation.

To bypass these issues, the method is rewritten as a true explicit one-step map, often called Velocity Verlet. By tracking an auxiliary velocity variable defined at the half-step, we propagate $(q_n, v_n) \mapsto (q_{n+1}, v_{n+1})$ directly:

$$v_{n+1/2} = v_n + \frac{h}{2}f(q_n) \quad (53)$$

$$q_{n+1} = q_n + hv_{n+1/2} \quad (54)$$

$$v_{n+1} = v_{n+1/2} + \frac{h}{2}f(q_{n+1}) \quad (55)$$

This formulation tracks both positions and velocities cleanly at the end of every time step and naturally accommodates variable step sizes.

7 The Adjoint of a Method and Symmetry

To mathematically prove the properties of the Störmer-Verlet scheme, we must first establish the concept of the adjoint of a numerical method. In classical physics, conservative systems exhibit perfect time-reversibility: if a system evolves forward in time by t and then backward by t , it returns exactly to its initial state. Mathematically, if φ_t represents the exact flow of an autonomous differential system $\dot{y} = f(y)$, it satisfies the property $\varphi_{-t}^{-1} = \varphi_t$, or equivalently, $\varphi_{-t} \circ \varphi_t = I$.

However, standard numerical integrators generally do not preserve this reversibility. For instance, executing a step of size h with the Explicit Euler method and then reversing it with a step of size $-h$ using the same formula does not return the system to its starting point. To formalize this behavior, we define the adjoint method.

Definition 3.1. *The adjoint method Φ_h^* of a numerical method Φ_h is the inverse map of the original method executed with a reversed time step $-h$:*

$$\Phi_h^* := \Phi_{-h}^{-1} \quad (56)$$

In practical terms, calculating a forward step $y_1 = \Phi_h^(y_0)$ with the adjoint method is implicitly defined by running the original method backward from y_1 to recover the initial state: $\Phi_{-h}(y_1) = y_0$.*

Example: The Adjoint of the Symplectic Euler Method

To illustrate how the adjoint mapping is explicitly calculated, let Φ_h represent the standard Symplectic Euler method:

$$p_1 = p_0 - h \frac{\partial H}{\partial q}(p_1, q_0) \quad (57)$$

$$q_1 = q_0 + h \frac{\partial H}{\partial p}(p_1, q_0) \quad (58)$$

To find its adjoint, we set up the inverse step Φ_{-h} . We apply the original method Φ to the state (q_1, p_1) with a step size of $-h$, requiring that the output is precisely the initial state (q_0, p_0) . To do this, we perform the following substitutions in our original equations:

- Initial state: $(q_0, p_0) \longrightarrow (q_1, p_1)$
- Final state: $(q_1, p_1) \longrightarrow (q_0, p_0)$
- Step size: $h \longrightarrow -h$

Applying these substitutions yields the backward step equation:

$$p_0 = p_1 - (-h) \frac{\partial H}{\partial q}(p_0, q_1) \quad (59)$$

$$q_0 = q_1 + (-h) \frac{\partial H}{\partial p}(p_0, q_1) \quad (60)$$

To write this as an explicit one-step method propagating forward from (q_0, p_0) to (q_1, p_1) , we simplify the double negatives and rearrange algebraically:

$$p_0 = p_1 + h \frac{\partial H}{\partial q}(p_0, q_1) \quad (61)$$

$$q_0 = q_1 - h \frac{\partial H}{\partial p}(p_0, q_1) \quad (62)$$

Isolating the target final state variables p_1 and q_1 , we obtain the mapped system representing the adjoint method Φ_h^* :

$$p_1 = p_0 - h \frac{\partial H}{\partial q}(p_0, q_1) \quad (63)$$

$$q_1 = q_0 + h \frac{\partial H}{\partial p}(p_0, q_1) \quad (64)$$

The adjoint operator adheres to several clean algebraic rules. Taking the adjoint of an adjoint returns the original method, meaning $(\Phi_h^*)^* = \Phi_h$. Furthermore, the adjoint of a composed sequence of methods reverses their order, perfectly mirroring matrix transposition: $(\Phi_h \circ \Psi_h)^* = \Psi_h^* \circ \Phi_h^*$.

These properties allow us to rigorously define numerical symmetry. A numerical method is called symmetric (or self-adjoint) if it is mathematically identical to its own adjoint:

$$\Phi_h^* = \Phi_h \quad (65)$$

While methods like Explicit Euler are not symmetric (its adjoint is the Implicit Euler method), integrators like the Implicit Midpoint Rule and the Störmer-Verlet scheme are strictly symmetric.

Theorem 3.2. *Let Φ_h be a one-step method of order p with a local error expansion given by:*

$$\Phi_h(y_0) = \varphi_h(y_0) + C(y_0)h^{p+1} + \mathcal{O}(h^{p+2}) \quad (66)$$

where φ_h is the exact flow of the differential equation and $C(y_0)$ is the principal error coefficient. Then, the adjoint method Φ_h^ has the exact same order p , and its local error expansion is:*

$$\Phi_h^*(y_0) = \varphi_h(y_0) + (-1)^p C(y_0)h^{p+1} + \mathcal{O}(h^{p+2}) \quad (67)$$

Furthermore, if the method is symmetric ($\Phi_h = \Phi_h^$), its maximal order is strictly even.*

Proof. The proof relies on a geometric construction known as "back-projection". Let us start at an initial state y_0 and define two final points after a forward step of size h :

1. The exact flow point: $y_{\text{exact}} = \varphi_h(y_0)$.
2. The adjoint step point: $y_1 = \Phi_h^*(y_0)$.

The difference between these two points is the local truncation error of the adjoint method, which we denote as e^* :

$$e^* = y_1 - \varphi_h(y_0) \quad (68)$$

To evaluate this, we "project" both of these final points back to the starting time by applying the original numerical method with a negative step size, $-h$. By the fundamental definition of the adjoint method ($\Phi_h^* := \Phi_{-h}^{-1}$), running the negative step from the adjoint point y_1 lands us exactly back at our starting point:

$$\Phi_{-h}(y_1) = y_0 \quad (69)$$

Conversely, running the negative step from the exact flow point y_{exact} lands us at a back-projected point $\Phi_{-h}(\varphi_h(y_0))$. The difference between these two back-projected points defines the back-projected local error, e :

$$e = y_0 - \Phi_{-h}(\varphi_h(y_0)) \quad (70)$$

Because the exact flow of a differential equation is perfectly time-reversible, we know that $y_0 = \varphi_{-h}(\varphi_h(y_0))$. Substituting this into our expression for e yields:

$$e = \varphi_{-h}(\varphi_h(y_0)) - \Phi_{-h}(\varphi_h(y_0)) \quad (71)$$

Notice that e is exactly the negative of the local error of the original method Φ taking a negative step size $-h$, starting from the exact point $\bar{y} = \varphi_h(y_0)$. Using our known local error expansion formula for Φ_h , we substitute $-h$ for the step size:

$$e = -(\Phi_{-h}(\bar{y}) - \varphi_{-h}(\bar{y})) = -(C(\bar{y})(-h)^{p+1} + \mathcal{O}((-h)^{p+2})) \quad (72)$$

Factoring out the negative sign gives:

$$e = (-1)^p C(\varphi_h(y_0))h^{p+1} + \mathcal{O}(h^{p+2}) \quad (3.5) \quad (73)$$

Deriving the Jacobian of the Step Map

Every standard, consistent numerical one-step method $\Phi_h(y)$ can be expressed in terms of an increment function $\psi(y, h)$:

$$\Phi_h(y) = y + h\psi(y, h) \quad (74)$$

For the method to be consistent with the differential equation $\dot{y} = f(y)$, the increment function must satisfy $\psi(y, 0) = f(y)$. Since the vector field $f(y)$ is smooth, the increment function $\psi(y, h)$ is continuously differentiable (C^1) with respect to the state variable y .

To compute the Jacobian matrix of this step map, we take the partial derivative of $\Phi_h(y)$ with respect to y :

$$\frac{\partial \Phi_h(y)}{\partial y} = \frac{\partial}{\partial y} [y + h\psi(y, h)] = \frac{\partial y}{\partial y} + \frac{\partial}{\partial y} [h\psi(y, h)] \quad (75)$$

The derivative of the state vector with respect to itself is the identity matrix I . Factoring out the scalar step size h yields:

$$\frac{\partial \Phi_h(y)}{\partial y} = I + h \frac{\partial \psi(y, h)}{\partial y} \quad (76)$$

Because $\psi(y, h)$ is C^1 , its Jacobian matrix $\frac{\partial \psi}{\partial y}$ is bounded in norm by some real constant M on any compact local neighbourhood of y :

$$\left\| \frac{\partial \psi(y, h)}{\partial y} \right\| \leq M \quad (77)$$

By the definition of asymptotic big- $\mathcal{O}$ notation, any matrix-valued function scaled by h whose norm is bounded by a constant times h is of order $\mathcal{O}(h)$:

$$h \frac{\partial \psi(y, h)}{\partial y} = \mathcal{O}(h) \quad (78)$$

Substituting this back into the Jacobian expression confirms that for any step size of order $\mathcal{O}(h)$ (including the negative step $-h$):

$$\frac{\partial \Phi_{-h}(y)}{\partial y} = I + \mathcal{O}(h) \quad (79)$$

Relating the Back-Projected Error e to the Forward Error e^*

Because the numerical step map Φ_{-h} is smooth and differentiable, we apply a multivariable Taylor expansion to relate the difference in its outputs to the difference in its inputs:

$$\Phi_{-h}(y_1) - \Phi_{-h}(\varphi_h(y_0)) = \frac{\partial \Phi_{-h}}{\partial y} \cdot (y_1 - \varphi_h(y_0)) + \mathcal{O}(\|y_1 - \varphi_h(y_0)\|^2) \quad (80)$$

Recalling the definition of our points, we have $\Phi_{-h}(y_1) = y_0$, which makes the left-hand side precisely our back-projected local error $e = y_0 - \Phi_{-h}(\varphi_h(y_0))$. Similarly, the term in parentheses on the right-hand side is our forward local error $e^* = y_1 - \varphi_h(y_0)$.

Substituting the Jacobian $\frac{\partial \Phi_{-h}}{\partial y} = I + \mathcal{O}(h)$ and noting that the quadratic error term $\mathcal{O}(\|e^*\|^2) = \mathcal{O}(h^{2p+2})$ is strictly absorbed into higher orders, we obtain:

$$e = (I + \mathcal{O}(h))e^* \quad (81)$$

Inverting this matrix relation yields:

$$e^* = (I + \mathcal{O}(h))^{-1}e = (I + \mathcal{O}(h))e \quad (82)$$

Evaluating the Principal Error Coefficient at y_0

From the previous stage of the proof, taking a negative step size $-h$ from the exact forward point $\varphi_h(y_0)$ yields a back-projected error of:

$$e = (-1)^p C(\varphi_h(y_0))h^{p+1} + \mathcal{O}(h^{p+2}) \quad (83)$$

However, we require the final formula for the adjoint error e^* to be evaluated at the initial state y_0 . Because the exact flow over a time interval h moves a distance proportional to the step size, we have $\varphi_h(y_0) = y_0 + \mathcal{O}(h)$.

Since the principal error coefficient function $C(y)$ is continuous and smooth, we evaluate it at this shifted point via Taylor expansion:

$$C(\varphi_h(y_0)) = C(y_0 + \mathcal{O}(h)) = C(y_0) + \mathcal{O}(h) \quad (84)$$

Substituting this back into our expression for e gives:

$$e = (-1)^p [C(y_0) + \mathcal{O}(h)] h^{p+1} + \mathcal{O}(h^{p+2}) \quad (85)$$

Distributing the h^{p+1} factor:

$$e = (-1)^p C(y_0)h^{p+1} + \mathcal{O}(h^{p+2}) + \mathcal{O}(h^{p+2}) = (-1)^p C(y_0)h^{p+1} + \mathcal{O}(h^{p+2}) \quad (86)$$

Analyzing the Higher-Order Terms for e^*

Finally, we substitute this simplified expression for e back into our inverted Jacobian relation:

$$e^* = (I + \mathcal{O}(h)) [(-1)^p C(y_0)h^{p+1} + \mathcal{O}(h^{p+2})] \quad (87)$$

Multiplying this out term-by-term:

$$e^* = I \cdot [(-1)^p C(y_0)h^{p+1}] + I \cdot \mathcal{O}(h^{p+2}) + \mathcal{O}(h) \cdot [(-1)^p C(y_0)h^{p+1}] + \mathcal{O}(h) \cdot \mathcal{O}(h^{p+2}) \quad (88)$$

We analyze the order of each component as $h \rightarrow 0$:

1. **First Term:** The leading-order error contribution, $(-1)^p C(y_0)h^{p+1}$.

2. **Second Term:** The primary remainder of order $\mathcal{O}(h^{p+2})$.
3. **Third Term:** Since $\mathcal{O}(h) \cdot h^{p+1} = \mathcal{O}(h^{p+2})$, this term is of order $\mathcal{O}(h^{p+2})$.
4. **Fourth Term:** Since $\mathcal{O}(h) \cdot \mathcal{O}(h^{p+2}) = \mathcal{O}(h^{p+3})$, this term decays even faster.

Since the second, third, and fourth terms are all of order $\mathcal{O}(h^{p+2})$ or higher, they collapse into a single remainder bucket:

$$e^* = (-1)^p C(y_0) h^{p+1} + \mathcal{O}(h^{p+2}) \quad (89)$$

This perfectly matches the required local error expansion of the adjoint method, completing the first part of the proof. $\square$

Corollary Proof (Even-Order Rule for Symmetric Methods)

If a numerical method is symmetric, it is equal to its own adjoint: $\Phi_h = \Phi_h^*$. Consequently, their local truncation error expansions must be identical. Equating their leading-order coefficients yields:

$$C(y_0) = (-1)^p C(y_0) \quad (90)$$

If the method's order p is odd, then $(-1)^p = -1$, which algebraically forces:

$$C(y_0) = -C(y_0) \implies 2C(y_0) = 0 \implies C(y_0) = 0 \quad (91)$$

Because the principal error coefficient $C(y_0)$ is forced to be exactly zero, the leading error term vanishes completely. The actual local error reduces to at least $\mathcal{O}(h^{p+2})$, which guarantees the physical order of accuracy jumps to at least $p+1$ (an even integer). Thus, the maximal order of a symmetric method must be even. $\square$

Theorem. *If a numerical mapping Φ_h is symplectic for any step size h , its adjoint method $\Phi_h^* = \Phi_{-h}^{-1}$ is also strictly symplectic.*

Proof. The proof leverages the algebraic group properties of symplectic matrices applied to the Jacobians of the mappings.

Step 1: Symplecticity of the Negative Step

By definition, if the numerical method Φ_h is symplectic, its Jacobian matrix $G = \Phi'_h(y)$ satisfies the symplecticity condition for any step size h :

$$G^T J G = J \quad (92)$$

Since $-h$ is simply a negative step size, the negative-step map Φ_{-h} must also be a valid symplectic transformation:

$$(\Phi'_{-h})^T J (\Phi'_{-h}) = J \quad (93)$$

Step 2: Symplecticity of the Inverse (The Jacobian Level)

Let $M = \Phi'_{-h}(y)$ be the Jacobian of our negative-step map. Since it is symplectic, we have:

$$M^T J M = J \quad (94)$$

By the inverse function theorem, the Jacobian of the inverse map Φ_{-h}^{-1} (which represents our adjoint method Φ_h^*) is simply the matrix inverse M^{-1} . To prove the adjoint is symplectic, we must show that this inverse Jacobian is also symplectic:

$$(M^{-1})^T J M^{-1} = J \quad (95)$$

We start with our known symplecticity condition for the forward Jacobian:

$$M^T J M = J \quad (96)$$

Since M is invertible, we can multiply on the left by the inverse transpose $(M^T)^{-1}$ and on the right by the inverse M^{-1} :

$$(M^T)^{-1} (M^T J M) M^{-1} = (M^T)^{-1} J M^{-1} \quad (97)$$

Using the identity matrix properties on the left-hand side, the terms cancel out:

$$I \cdot J \cdot I = (M^T)^{-1} J M^{-1} \implies J = (M^T)^{-1} J M^{-1} \quad (98)$$

Since the matrix operations of transpose and inversion commute—meaning $(M^T)^{-1} = (M^{-1})^T$ —we substitute this into our equation to get:

$$(M^{-1})^T J M^{-1} = J \quad (99)$$

This mathematically proves that the Jacobian of the inverse map M^{-1} is strictly symplectic. Consequently, the adjoint of any symplectic method is guaranteed to preserve symplecticity. $\square$

8 Composition Methods

The composition method is an elegant and powerful technique designed to construct high-order numerical integrators from simple, low-order basic methods. A primary advantage of this approach is that if the underlying basic method preserves a specific geometric structure—such as symplecticity—the resulting composed method is mathematically guaranteed to preserve that exact same structure.

Suppose we are given a basic one-step numerical method Φ_h of order p . Instead of advancing the system by a single step of size h , we split the step into s fractional sub-steps of sizes $\gamma_1 h, \gamma_2 h, \dots, \gamma_s h$. We apply the basic method sequentially using these sub-step sizes to define the corresponding composition method Ψ_h :

$$\Psi_h = \Phi_{\gamma_s h} \circ \dots \circ \Phi_{\gamma_2 h} \circ \Phi_{\gamma_1 h} \quad (100)$$

Theorem 4.1. *Let Φ_h be a one-step method of order p . If the fractional step coefficients γ_i satisfy the consistency and order-increase conditions:*

$$\gamma_1 + \gamma_2 + \cdots + \gamma_s = 1 \quad (101)$$

$$\gamma_1^{p+1} + \gamma_2^{p+1} + \cdots + \gamma_s^{p+1} = 0 \quad (102)$$

then the composition method Ψ_h is at least of order $p + 1$.

Proof. We analyze the global error of the composed method by tracking how the leading local truncation error accumulates and propagates across each of the s sub-steps.

Step 1: Defining the Local Stage Errors

Let y_0 be the initial state and let y_{i-1} be the numerical state before the i -th sub-step. The local stage error e_i is the error committed strictly during the i -th sub-step of size $\gamma_i h$, defined as the difference between the numerical step and the exact flow $\varphi_{\gamma_i h}$ starting from the same intermediate point:

$$e_i = \Phi_{\gamma_i h}(y_{i-1}) - \varphi_{\gamma_i h}(y_{i-1}) \quad (103)$$

Because the basic method Φ is of order p , this local error is characterized by a principal error coefficient function $C(y)$:

$$e_i = C(y_{i-1})(\gamma_i h)^{p+1} + \mathcal{O}(h^{p+2}) \quad (104)$$

Step 2: Evaluating the Error Coefficient at y_0

Because each sub-step is of size $\mathcal{O}(h)$, the intermediate state variables remain close to the initial state, satisfying $y_{i-1} = y_0 + \mathcal{O}(h)$. Assuming the continuous error coefficient function $C(y)$ is smooth, we can expand it around y_0 using a Taylor series:

$$C(y_{i-1}) = C(y_0 + \mathcal{O}(h)) = C(y_0) + \mathcal{O}(h) \quad (105)$$

Substituting this expansion back into our expression for the local stage error yields:

$$e_i = (C(y_0) + \mathcal{O}(h))\gamma_i^{p+1}h^{p+1} + \mathcal{O}(h^{p+2}) = C(y_0)\gamma_i^{p+1}h^{p+1} + \mathcal{O}(h^{p+2}) \quad (106)$$

Step 3: Propagating Stage Errors to the Final Time

The local error e_i committed at stage i does not remain static; it propagates under the exact flow of the differential equation for the remainder of the full time step. Let E_i denote this propagated error evaluated at the final time $t = h$.

As established in the analysis of flow maps, the exact flow over a small time interval behaves as the identity mapping plus a perturbation of order $\mathcal{O}(h)$. Consequently, the Jacobian matrix that propagates this small discrepancy forward is $I + \mathcal{O}(h)$. Therefore, the propagated stage error relates to the local stage error by:

$$E_i = (I + \mathcal{O}(h))e_i \quad (107)$$

Step 4: Summing the Telescoping Errors

The total global error of the composed step is the difference between the exact

flow over the full step and the numerical composed step. By expressing this as a telescoping sum of the individual propagated stage errors (Lady Windermere's Fan), we obtain:

$$\varphi_h(y_0) - \Psi_h(y_0) = \sum_{i=1}^s E_i = \sum_{i=1}^s (I + \mathcal{O}(h))e_i \quad (108)$$

Substituting our expanded expression for e_i into the sum:

$$\varphi_h(y_0) - \Psi_h(y_0) = \sum_{i=1}^s (I + \mathcal{O}(h)) \left(C(y_0) \gamma_i^{p+1} h^{p+1} + \mathcal{O}(h^{p+2}) \right) \quad (109)$$

Multiplying this out, any term containing $\mathcal{O}(h) \cdot h^{p+1}$ collapses into the $\mathcal{O}(h^{p+2})$ remainder bucket. We can then factor out the common leading-order components:

$$\varphi_h(y_0) - \Psi_h(y_0) = C(y_0) \left(\sum_{i=1}^s \gamma_i^{p+1} \right) h^{p+1} + \mathcal{O}(h^{p+2}) \quad (110)$$

Conclusion

By enforcing the algebraic condition of the theorem, $\sum_{i=1}^s \gamma_i^{p+1} = 0$, the entire leading error term strictly vanishes. Consequently, the total error of the composed method Ψ_h reduces to:

$$\varphi_h(y_0) - \Psi_h(y_0) = \mathcal{O}(h^{p+2}) \quad (111)$$

This mathematically proves that the composed method achieves an order of at least $p + 1$. $\square$

Theorem 3.6. *Let Φ_h be a symplectic numerical method. Then, any composition method defined by*

$$\Psi_h = \Phi_{\gamma_s h} \circ \cdots \circ \Phi_{\gamma_1 h} \quad (112)$$

remains strictly symplectic for any arbitrary choice of real step coefficients $\gamma_1, \dots, \gamma_s$.

Proof. The proof relies on the algebraic property that symplectic transformations form a mathematical group under function composition. We demonstrate this by tracking the Jacobian of composed mappings.

Step 1: The Definition of a Symplectic Mapping

Recall that a differentiable mapping $g : U \rightarrow \mathbb{R}^{2d}$ is symplectic if and only if its Jacobian matrix $g'(y)$ satisfies the symplecticity condition everywhere on its domain:

$$g'(y)^T J g'(y) = J \quad (113)$$

where J is the standard skew-symmetric structure matrix.

Step 2: Composition of Two Symplectic Mappings

Let g_1 and g_2 be two arbitrary symplectic mappings. We evaluate the symplecticity of their composition $G = g_2 \circ g_1$. By the chain rule of multivariable

calculus, the Jacobian of the composed map $G(y) = g_2(g_1(y))$ is the product of their individual Jacobians:

$$G'(y) = g'_2(g_1(y)) \cdot g'_1(y) \quad (114)$$

We test if this composed Jacobian satisfies the definition of symplecticity:

$$G'(y)^T J G'(y) = (g'_2(g_1(y))g'_1(y))^T J (g'_2(g_1(y))g'_1(y)) \quad (115)$$

Using the matrix transpose identity $(AB)^T = B^T A^T$, we expand the right-hand side:

$$G'(y)^T J G'(y) = g'_1(y)^T [g'_2(g_1(y))^T J g'_2(g_1(y))] g'_1(y) \quad (116)$$

Because g_2 is a symplectic mapping, its Jacobian satisfies $g'_2(z)^T J g'_2(z) = J$ for any point z in its domain. Substituting $z = g_1(y)$ collapses the inner bracketed term directly to J :

$$G'(y)^T J G'(y) = g'_1(y)^T J g'_1(y) \quad (117)$$

Similarly, because g_1 is also symplectic, its Jacobian satisfies $g'_1(y)^T J g'_1(y) = J$. Substituting this yields:

$$G'(y)^T J G'(y) = J \quad (118)$$

This rigorously proves that the composition of two symplectic mappings is always strictly symplectic.

Step 3: Mathematical Induction

By induction, this algebraic property generalizes to the composition of any finite sequence of s symplectic mappings. If $g_1, g_2, \dots, g_s$ are all individual symplectic transformations, then their sequential composition $G = g_s \circ \dots \circ g_2 \circ g_1$ is guaranteed to satisfy $G'(y)^T J G'(y) = J$, making the final composed mapping strictly symplectic.

Step 4: Application to the Composition Method

In a general composition method, we define our composed one-step map Ψ_h by sequentially applying our basic method Φ_h using a set of step coefficients:

$$\Psi_h = \Phi_{\gamma_s h} \circ \dots \circ \Phi_{\gamma_1 h} \quad (119)$$

Because the basic method Φ_h is inherently symplectic, each individual sub-stage map $g_i = \Phi_{\gamma_i h}$ is a perfectly valid symplectic mapping (it is simply the basic symplectic method evaluated with an individual step size of $h_i = \gamma_i h$). Since each sub-stage is symplectic, their composition Ψ_h must be strictly and unconditionally symplectic. $\square$

Composition with the Adjoint Method

In our standard composition method, we encountered a fundamental algebraic bottleneck: if the basic method's order p is odd, the condition $\sum \gamma_i^{p+1} = 0$ has no real, non-zero solutions because $p+1$ is even, making every term strictly positive. To achieve an order increase for methods with odd p , we must generalize our approach by explicitly interleaving the basic method Φ_h with its adjoint Φ_h^* .

We replace the standard composition formula with a more general sequence utilizing two sets of step coefficients, α_i and β_i :

$$\Psi_h = \Phi_{\alpha_s h} \circ \Phi_{\beta_s h}^* \circ \cdots \circ \Phi_{\alpha_2 h} \circ \Phi_{\beta_2 h}^* \circ \Phi_{\alpha_1 h} \circ \Phi_{\beta_1 h}^* \quad (120)$$

To determine the conditions required for this composed method Ψ_h to achieve order $p+1$, we rely on Theorem 3.2, which establishes that the leading local error term of the adjoint method is scaled by $(-1)^p$. By accumulating the propagated stage errors as we did previously, the condition for order $p+1$ becomes:

$$\beta_1 + \alpha_1 + \beta_2 + \alpha_2 + \cdots + \beta_s + \alpha_s = 1 \quad (121)$$

$$(-1)^p \beta_1^{p+1} + \alpha_1^{p+1} + (-1)^p \beta_2^{p+1} + \alpha_2^{p+1} + \cdots + (-1)^p \beta_s^{p+1} + \alpha_s^{p+1} = 0 \quad (122)$$

Because of the alternating $(-1)^p$ factor introduced by the adjoint steps, the error terms can successfully cancel one another out even when p is odd, entirely resolving the previous real-solution bottleneck.

The $s = 1$ Symmetric Construction

A particularly powerful and immediate application of this generalized formula occurs for a single stage ($s = 1$) starting from a consistent, first-order method ($p = 1$). The order conditions simplify to:

$$\beta_1 + \alpha_1 = 1 \quad (123)$$

$$-\beta_1^2 + \alpha_1^2 = 0 \quad (124)$$

We can see at once that the solution is $\alpha_1 = \beta_1 = 1/2$. This implies that every consistent one-step method of order 1 can be automatically transformed into a second-order, symmetric method via the composition:

$$\Psi_h = \Phi_{h/2} \circ \Phi_{h/2}^* \quad (125)$$

9 The General Störmer-Verlet Method for Hamiltonian Systems

Initially, we discussed the Störmer-Verlet method in the context of a simple second-order differential equation. However, for an arbitrary, fully coupled, non-separable Hamiltonian $H(p, q)$, the Störmer-Verlet method is formally defined precisely as the half-step composition of the Symplectic Euler method with its adjoint:

$$\Psi_h = \Phi_{h/2} \circ \Phi_{h/2}^* \quad (126)$$

Without making any assumptions about the structure of $H(p, q)$, this composition yields a 4-step implicit algorithm to advance the system from $(q_n, p_n) \rightarrow (q_{n+1}, p_{n+1})$.

Step 1: The Adjoint Half-Step $\Phi_{h/2}^*$

Using the definition of the adjoint of Symplectic Euler, we solve the implicit equations for the intermediate coordinates $(q_{n+1/2}, p_{n+1/2})$:

$$q_{n+1/2} = q_n + \frac{h}{2} \frac{\partial H}{\partial p}(p_n, q_{n+1/2}) \quad (127)$$

$$p_{n+1/2} = p_n - \frac{h}{2} \frac{\partial H}{\partial q}(p_n, q_{n+1/2}) \quad (128)$$

Step 2: The Symplectic Euler Half-Step $\Phi_{h/2}$

Using the standard Symplectic Euler equations, we complete the step to reach (q_{n+1}, p_{n+1}) :

$$p_{n+1} = p_{n+1/2} - \frac{h}{2} \frac{\partial H}{\partial q}(p_{n+1}, q_{n+1/2}) \quad (129)$$

$$q_{n+1} = q_{n+1/2} + \frac{h}{2} \frac{\partial H}{\partial p}(p_{n+1}, q_{n+1/2}) \quad (130)$$

The Explicit Formulation for Separable Hamiltonians

For a single stage ($s = 1$) starting from a first-order method ($p = 1$), the consistency and order conditions yield a symmetric composition with half-step sizes: $\Psi_h = \Phi_{h/2} \circ \Phi_{h/2}^*$. We can prove algebraically that this composition corresponds exactly to the canonical Störmer-Verlet scheme for a separable physical Hamiltonian of the form $H(p, q) = T(p) + U(q)$.

Step 1: Apply the Adjoint Method with step size $h/2$

Starting from (q_n, p_n) , we compute the intermediate half-step state $(q_{n+1/2}, p_{n+1/2}) = \Phi_{h/2}^*(q_n, p_n)$:

$$q_{n+1/2} = q_n + \frac{h}{2} \frac{\partial H}{\partial p}(p_n, q_{n+1/2}) = q_n + \frac{h}{2} \nabla T(p_n) \quad (131)$$

$$p_{n+1/2} = p_n - \frac{h}{2} \frac{\partial H}{\partial q}(p_n, q_{n+1/2}) = p_n - \frac{h}{2} \nabla U(q_{n+1/2}) \quad (132)$$

Step 2: Apply the Original Method with step size $h/2$

Starting from $(q_{n+1/2}, p_{n+1/2})$, we complete the full step $(q_{n+1}, p_{n+1}) = \Phi_{h/2}(q_{n+1/2}, p_{n+1/2})$:

$$p_{n+1} = p_{n+1/2} - \frac{h}{2} \frac{\partial H}{\partial q}(p_{n+1}, q_{n+1/2}) = p_{n+1/2} - \frac{h}{2} \nabla U(q_{n+1/2}) \quad (133)$$

$$q_{n+1} = q_{n+1/2} + \frac{h}{2} \frac{\partial H}{\partial p}(p_{n+1}, q_{n+1/2}) = q_{n+1/2} + \frac{h}{2} \nabla T(p_{n+1}) \quad (134)$$

Step 3: Eliminating the Intermediate Momentum $p_{n+1/2}$

Substituting the equation for $p_{n+1/2}$ directly into the update equation for p_{n+1} yields:

$$p_{n+1} = \left[p_n - \frac{h}{2} \nabla U(q_{n+1/2}) \right] - \frac{h}{2} \nabla U(q_{n+1/2}) = p_n - h \nabla U(q_{n+1/2}) \quad (135)$$

Gathering these steps in sequence gives the nice explicit **Position-Verlet** formulation of the Störmer-Verlet scheme:

$$q_{n+1/2} = q_n + \frac{h}{2} \nabla T(p_n) \quad (136)$$

$$p_{n+1} = p_n - h \nabla U(q_{n+1/2}) \quad (137)$$

$$q_{n+1} = q_{n+1/2} + \frac{h}{2} \nabla T(p_{n+1}) \quad (138)$$

If we compose the methods in the reverse order ($\Psi_h = \Phi_{h/2}^* \circ \Phi_{h/2}$), we recover the standard explicit **Velocity-Verlet (Momentum-Verlet)** formulation:

$$p_{n+1/2} = p_n - \frac{h}{2} \nabla U(q_n) \quad (139)$$

$$q_{n+1} = q_n + h \nabla T(p_{n+1/2}) \quad (140)$$

$$p_{n+1} = p_{n+1/2} - \frac{h}{2} \nabla U(q_{n+1}) \quad (141)$$

Theorem 3.5. *The Störmer-Verlet scheme is a symplectic numerical method of order 2.*

Proof. We prove this using the mathematical properties of composition and adjoint mappings established previously in this report.

Part A: Proof of Symplecticity

1. **Symplecticity of the Base Method:** By Theorem 3.3, the Symplectic Euler method Φ_h is a symplectic map for any smooth Hamiltonian system.
2. **Symplecticity of the Adjoint:** Since we have mathematically proven that the adjoint of any symplectic map is also symplectic, the Adjoint Symplectic Euler method Φ_h^* is unconditionally symplectic.
3. **Symplecticity of Composition:** By Theorem 3.6, any composition method of the form $\Psi_h = \Phi_{\gamma_s h} \circ \dots \circ \Phi_{\gamma_1 h}$, or any generalized composition utilizing the adjoint, is strictly symplectic for any choice of step coefficients if all the functions being composed are symplectic.
4. **Conclusion of Symplecticity:** Since the Störmer-Verlet scheme is defined precisely as the composition of two symplectic mappings, namely:

$$\Psi_h = \Phi_{h/2} \circ \Phi_{h/2}^* \quad (142)$$

it is mathematically guaranteed to be strictly symplectic.

Part B: Proof of Order 2 (Symmetry)

1. **The Transpose Rule of Composed Adjoints:** For any composed sequence, the adjoint operator reverses the order of composition:

$$(\Phi_h \circ \Psi_h)^* = \Psi_h^* \circ \Phi_h^* \quad (143)$$

2. **Symmetry of Störmer-Verlet:** We evaluate the adjoint of the complete Störmer-Verlet map Ψ_h :

$$\Psi_h^* = \left(\Phi_{h/2} \circ \Phi_{h/2}^* \right)^* = (\Phi_{h/2}^*)^* \circ \Phi_{h/2}^* \quad (144)$$

Using the double adjoint property, $(\Phi^*)^* = \Phi$, this expression simplifies to:

$$\Psi_h^* = \Phi_{h/2} \circ \Phi_{h/2}^* = \Psi_h \quad (145)$$

Since $\Psi_h^* = \Psi_h$, the Störmer-Verlet scheme is strictly symmetric (self-adjoint).

3. **The Even-Order Rule:** By Theorem 3.2, the maximal order of accuracy p of any symmetric method must be strictly even.
4. Since the base method is consistent (meaning $p \geq 1$), and the symmetry constraint analytically forces an even order, the order of the Störmer-Verlet method automatically jumps to order 2.

This completes the rigorous, structure-preserving proof of Theorem 3.5. $\square$

10 Numerical Experiments

10.1 Experiment 1: Phase Space Area Evolution of Symplectic and Non-Symplectic Methods

To empirically demonstrate the failure of standard numerical integrators to preserve the geometric structure of Hamiltonian systems, we track the evolution of a defined phase space area over time.

The background trajectories for the system are generated by selecting multiple initial states along the p -axis and evolving them until $t = 15$ using a step size of $h = 0.5$. These trajectories visually define the underlying flow of the differential equation.

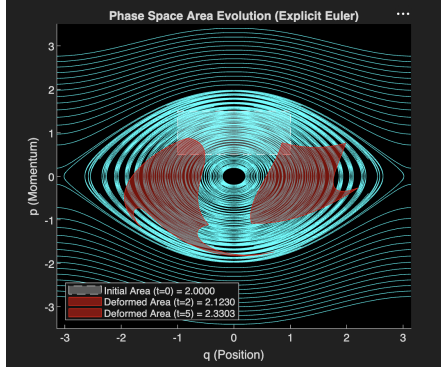
We define an initial rectangular region in the phase space spanning the position coordinates $q \in [-1, 1]$ and the momentum coordinates $p \in [0.5, 1.5]$. The exact initial oriented area of this region is 2.0000. To approximate the continuous manifold computationally, we discretize the boundary of this rectangle by placing 500 equidistant points along each of its four edges, forming a closed, 2000-sided polygon.

Using this 2000-point boundary as our set of initial conditions, we propagate the system forward in time using four distinct, non-symplectic Runge-Kutta methods: Explicit Euler (order 1), Heun's Method (order 2), Standard RK3 (order 3), and Standard RK4 (order 4). We record the deformed boundary coordinates at $t = 2$ and $t = 5$. By applying the standard polygon area formula to these advected boundary points, we approximate the area of the deformed region. This calculation inherently assumes that the topological mapping is continuous, meaning that all points initially contained within the boundary remain strictly enclosed by the deformed boundary.

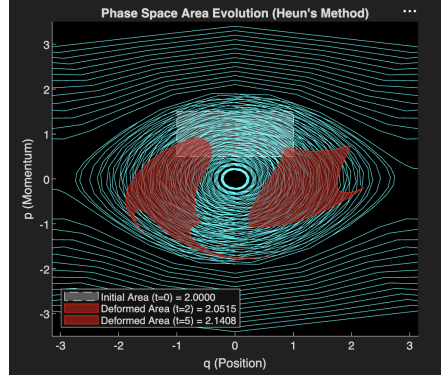
The experimental results explicitly confirm that non-symplectic methods do not conserve phase space area:

- Under the **Explicit Euler** method, the area artificially expands to 2.1230 at $t = 2$ and 2.3303 at $t = 5$.
- Under **Heun's Method**, the area expands to 2.0515 at $t = 2$ and 2.1408 at $t = 5$.
- Under **Standard RK3**, the area artificially dissipates to 1.9815 at $t = 2$ and 1.9515 at $t = 5$.
- Under **Standard RK4**, the area dissipates slightly to 1.9991 at $t = 2$ and 1.9976 at $t = 5$.

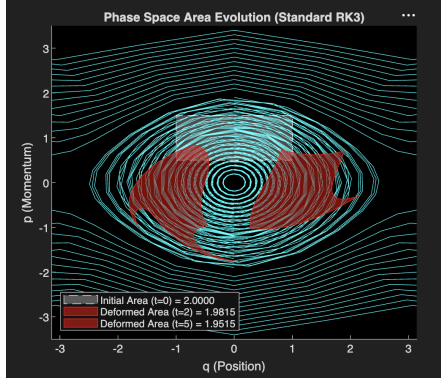
While none of these methods conserve the geometric area exactly, the magnitude of the area deviation from the true initial area monotonically decreases as the formal order of the numerical method increases.



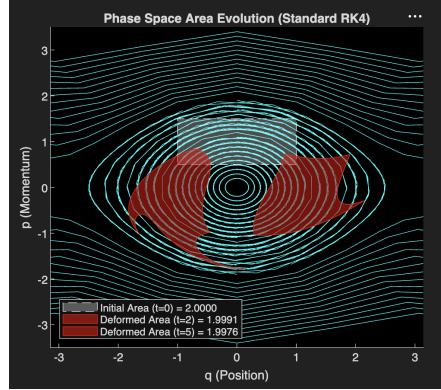
(a) Explicit Euler (Order 1)



(b) Heun's Method (Order 2)



(c) Standard RK3 (Order 3)



(d) Standard RK4 (Order 4)

Figure 1: Phase space area evolution of a rectangular initial domain under various non-symplectic methods at $t = 0$, $t = 2$, and $t = 5$.

To contrast this with structure-preserving algorithms, we repeat the exact same experimental setup—an initial rectangular domain of area 2.0000 discretized by 2000 boundary points, integrated with the same large step size of $h = 0.5$ —using two symplectic integrators: the Symplectic Euler method (order 1) and the Störmer-Verlet scheme (order 2).

Because we are using a relatively large step size, there are noticeable truncation errors in the exact placement and deformation of the macroscopic shapes compared to the true, continuous flow of the differential equation. The "tails" of the advected regions stretch and warp significantly over time.

However, the defining characteristic of these methods is their geometric structure preservation. Despite the visible trajectory deformations, both the Symplectic Euler and Störmer-Verlet methods strictly conserve the phase space area. For both methods, the calculated area of the advected 2000-sided polygon remains exactly 2.0000 at $t = 2$ and identically 2.0000 at $t = 5$.

This empirical result perfectly validates the theoretical proofs established in the preceding sections: symplectic integrators unconditionally preserve the geometric area of the phase space, entirely independent of the chosen step size or the method’s formal order of accuracy.

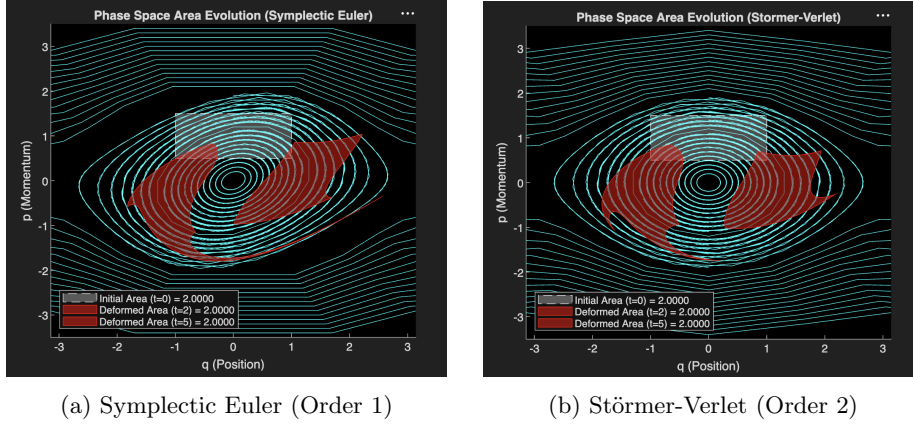


Figure 2: Phase space area evolution under symplectic numerical methods at $t = 0$, $t = 2$, and $t = 5$. Despite the shape deformation caused by the large step size ($h = 0.5$), the oriented area is perfectly conserved at 2.0000 throughout the simulation.

10.2 Experiment 2: Long-Term Area Conservation Analysis

In our second experiment, we evaluate the long-term structural stability of these numerical integrators by tracking the absolute error in the phase space area over an extended time domain, from $t = 0$ to $t = 100$.

The initial rectangular domain remains identical to the first experiment. However, to ensure high precision over the extended integration period, the boundary is discretized more finely into 4000 equidistant points. We execute the simulations using a step size of $h = 0.2$. To provide a rigorous geometric evaluation, we compare methods of equivalent formal order: the first-order Explicit Euler method against the first-order Symplectic Euler method, and the second-order Heun’s Method against the second-order Störmer-Verlet scheme.

The resulting absolute error plots demonstrate a profound divergence in long-term behavior between the non-symplectic and symplectic integrators:

- **First-Order Comparison:** The absolute area error of the non-symplectic Explicit Euler method diverges and oscillates wildly, exceeding an error magnitude of 350 over the integration period. Conversely, the Symplectic Euler method strictly preserves the phase space geometry, maintaining an absolute error that is consistently bounded near zero.

- **Second-Order Comparison:** Heun's Method, despite being a higher-order non-symplectic integrator, exhibits a steady, unbounded linear accumulation of area error over time. In stark contrast, the symplectic Störmer-Verlet scheme perfectly conserves the phase space volume, with its absolute error remaining entirely flat at zero for the duration of the simulation.

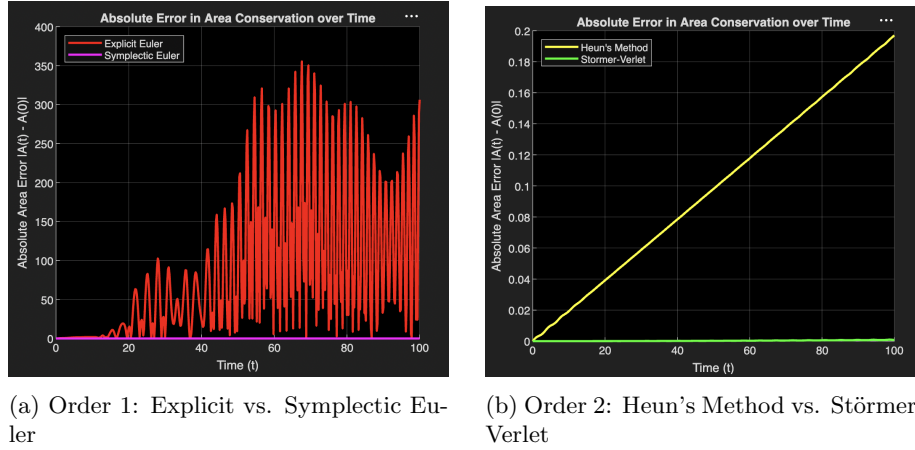


Figure 3: Absolute error in phase space area conservation over an extended time domain ($t \in [0, 100]$) with a step size of $h = 0.2$. Symplectic methods (Symplectic Euler and Störmer-Verlet) strictly maintain near-zero absolute error, while their non-symplectic counterparts display severe unbounded error growth.